Limit

Mind map

Existence

- LHL = RHL = finite
- $f(a-h)$, $f(a+h)$
- One side out of domain: other side

Indeterminate forms

- 7 forms; 0, 1 only approached
- Exact 0 or 1: determinate

Algebra of limits

- Both finite: combine
- Split only finite parts

Factorise / rationalise

- Cancel $(x-a)$
- Conjugate: square or cube
- $\infty - \infty$: LCM

Substitution

- $x = a + h$
- $x = \frac{1}{t}$ for ∞
- $x = -t$, $\sqrt{x^2} = -x$

Infinity/infinity

- Divide by highest power
- $\ln x \ll x^n \ll e^x$

Algebraic & trig limits

- $\frac{x^n - a^n}{x-a} \to na^{n-1}$
- $\frac{\sin x}{x} \to 1$, $\frac{1-\cos x}{x^2} \to \frac{1}{2}$
- Replace factors only

Exp & log limits

- $\frac{a^x - 1}{x} \to \ln a$
- $\frac{\ln(1+x)}{x} \to 1$

1 to the power infinity

- $e^{\lim g(f-1)}$
- Base $\to 1$, not exact

Expansions

- Compare lowest powers
- Four standard results
- Constants: lower coefficients = 0

Derivative limits

- $f'(a)$ as a limit
- f'': $(h-1)^2$ pattern

Special signs

- LHL, RHL separately
- $\{x\} \to 1$ from left of integer
- $\frac{\sin x}{x} \to 1^-$, $\frac{\tan x}{x} \to 1^+$

Sandwich theorem

- $x - 1 < [x] \leq x$
- Series: n·min, n·max

Composite & built-in

- Inner limit and its side
- $\lim x^n$: $|x| < 1, = 1, > 1$
- Continuous outer: limit inside

Special functions & geometry

- Rational/irrational: where definitions agree
- Areas in θ, then expand